

HDOC Day-5 Activity

- Question:

Find the product of two integers without using the multiplication operator *. The given algorithm and flowchart are incomplete. Correct them using repeated addition so that the logic uses the minimum possible number of additions and works for positive, negative and zero values.

- **Step 1: Understanding the Problem**

- **Input:** Two integers num1 and num2

- **Output:** Product of num1 and num2

- **Method:** Repeated addition without using the * operator

- **Requirement:** Use the smaller absolute value as the repetition count to obtain the minimum number of additions.

Mistakes in the Given Algorithm

- The loop condition uses $\text{index} \leq \text{num2}$ directly. If num2 is negative, the loop does not execute correctly.
- The algorithm always adds num1 exactly num2 times, so it does not guarantee the minimum number of additions.
- It does not handle positive $\times$ negative, negative $\times$ positive and negative $\times$ negative correctly.
- There is no separate handling for multiplication by zero.
- The sign of the final product is not determined separately from the repeated-addition process.

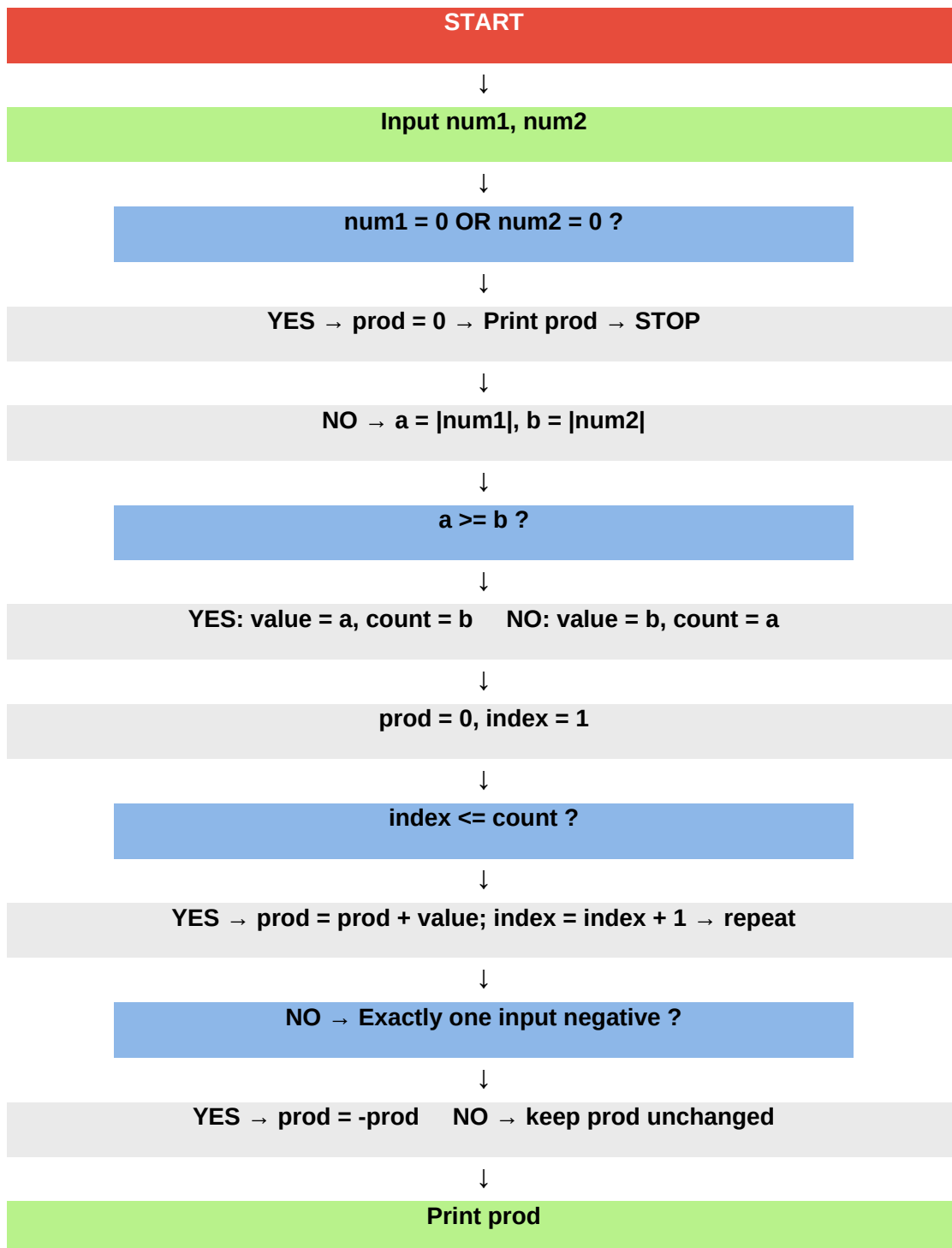
Corrected Algorithm

1. Start
2. Define num1, num2, a, b, value, count, prod, index as integers.
3. Input num1 and num2.
4. If num1 = 0 OR num2 = 0, set prod = 0 and go to Step 12.
5. Set a = absolute value of num1 and b = absolute value of num2.
6. If a $\geq$ b, set value = a and count = b; otherwise set value = b and count = a.
7. Set prod = 0 and index = 1.
8. Repeat while index $\leq$ count:
 - prod = prod + value
 - index = index + 1
9. If exactly one of num1 and num2 is negative, set prod = -prod.
10. Otherwise keep prod positive.
11. The number of additions used is count.
12. Print prod.
13. Stop.

Mistakes in the Given Flowchart

- The decision box checks $\text{index} \leq \text{num2}$, which fails when num2 is negative.
- The flowchart does not compare the absolute values of the two inputs to minimize additions.
- There is no branch for zero input.
- There is no logic to determine whether the result should be positive or negative.
- Therefore the original flowchart works only for limited positive-input cases.

Corrected Flowchart





STOP